GYANMANJARI INNOVATIVE UNIVERSITY

Gyanmanjari Institute of Technology – Faculty of Engineering &Technology

Data Mining Algorithm Implementation

Association rule mining with Apriori and FP-Growth on the RetailMart warehouse data

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Activity	ALA 2	Marks	10

Deliverables: the executed Jupyter notebook `DWDM_ALA2_Association_Rules_Notebook.ipynb` (code with outputs) and this summary of observations. The notebook reads the warehouse database created in ALA 1 (`retail_dw.sqlite`, supplied in the supporting-files archive).

1. Aim and Task

To select a dataset and implement one data mining task. The task chosen is **association rule mining** (market-basketanalysis).Twoalgorithmsareimplemented**fromscratch**–**Apriori**and**FP-Growth**–then the data are preprocessed, the algorithms are run and the results are evaluated with suitable metrics (support, confidence, lift, leverage, conviction).

2. Dataset and Preprocessing

Item	Description
Source	RetailMartdatawarehouse(ALA1):tablesfact_salesanddim_product.Thedataare**simulated** sample data, so patterns illustrate the method.
Transaction	One invoice = one basket; the products on the invoice are the items.
Size	5,153 invoice lines →1,499 invoices, 36 distinct products, mean basket size 3.44 items.
Preprocessing	Checked for missing values (none after the ETL in ALA 1); removed repeated items within an invoice;groupedlinesintobaskets;removed52single-itembaskets(norulecanbeformedfrom one item). **1,447 transactions** were mined.

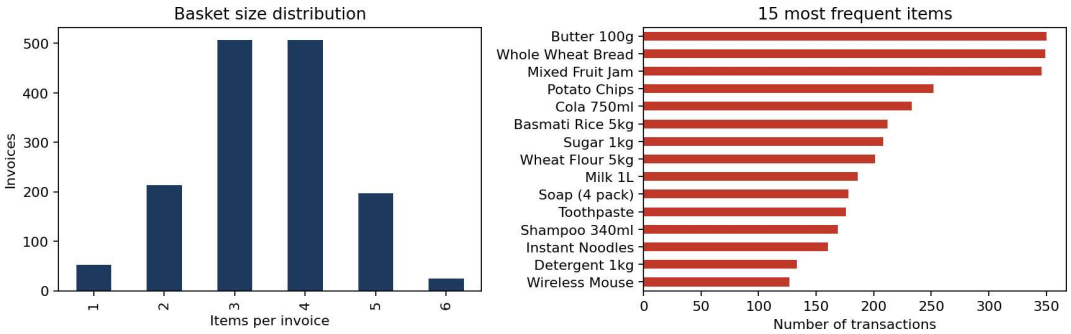


Figure 1: Basket-size distribution and the 15 most frequent items.

3. Algorithms Implemented

	Apriori	FP-Growth
Idea	Level-wisesearch:frequentk-itemsetsare joined to form (k+1)-candidates	CompressthedatabaseintoanFP-tree;mine it recursively with conditional pattern bases
Key principle	Apriori principle: all subsets of a frequent itemsetarefrequent→prunecandidateswith an infrequent subset	Itemsorderedbydescendingfrequencyso that transactions share prefixes in the tree
Candidate generation	Yes (join + prune)	No
Database scans	One per level (k scans)	Two (count items, build the tree)
Weakness	Many candidates; repeated scans	Treemaybelargefordensedata;more complex code

Rule generation: for each frequent itemset, every non-empty proper subset A is tried as an antecedent; the rule A →(itemset −A) is kept when its confidence is at least the threshold. Metrics: support = sup(A∪B); confidence = sup(A∪B)/sup(A); lift = confidence/sup(B); leverage = sup(A∪B) −sup(A)·sup(B); conviction = (1 −sup(B))/(1 −confidence).

4. Results

Frequent itemsets (minimum support 2%)

Both algorithms returned the same **128** frequent itemsets with identical supports; the supports of 25 randomly selected itemsets were also confirmed by direct counting in the database. By size: 1-itemsets: 36, 2-itemsets: 48, 3-itemsets: 33, 4-itemsets: 10, 5-itemsets: 1. Of these, 120 are *closed* (no superset with the same support) and 17 are *maximal* (no frequent superset).

Level k	Candidates generated (Apriori)	Frequent itemsets found
1	36	36
2	630	48
3	33	33
4	10	10
5	1	1

Most frequent itemsets (size ≥2)	Support
Butter 100g, Whole Wheat Bread	0.218
Mixed Fruit Jam, Whole Wheat Bread	0.214
Butter 100g, Mixed Fruit Jam	0.213
Butter 100g, Mixed Fruit Jam, Whole Wheat Bread	0.208
Cola 750ml, Potato Chips	0.136
Basmati Rice 5kg, Sugar 1kg	0.120
Basmati Rice 5kg, Wheat Flour 5kg	0.117
Sugar 1kg, Wheat Flour 5kg	0.116

Effect of the minimum support threshold; Apriori vs FP-Growth

Min support	Frequent itemsets	Apriori candidates	Apriori time (ms)	FP-Growthtime (ms)
1.0%	161	735	47.8	4.6
1.5%	142	718	46.7	4.4
2.0%	128	710	47.2	4.7
3.0%	101	595	38.9	4.2
5.0%	73	459	30.6	4.7

Best of 5 runs for each setting (Python, this machine).

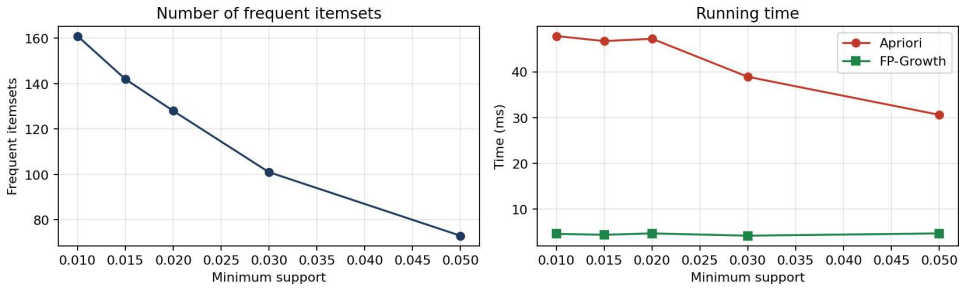


Figure 2: Number of frequent itemsets and running time against the minimum support.

Loweringthethresholdincreasesthenumberoffrequentitemsets(from73at5%to161at1%). FP-Growth was faster than Apriori in 5 of 5 settings, because it avoids repeated database scans and candidate testing. On such a small data set both run in milliseconds; the advantage matters for large data.

Association rules (support ≥2%, confidence ≥50%)

226 rules were generated; all have lift above 1. The ten rules with the highest lift:

Antecedent →Consequent	Support	Confidence	Lift
Fast Charger, Phone Case →Screen Guard, Smartphone	0.026	0.927	14.9
Fast Charger, Screen Guard →Phone Case, Smartphone	0.026	0.844	13.7
Fast Charger, Smartphone →Phone Case, Screen Guard	0.026	0.826	13.6
Fast Charger, Phone Case, Smartphone →Screen Guard	0.026	0.974	11.9
Phone Case, Smartphone →Screen Guard	0.059	0.966	11.8
Screen Guard →Phone Case, Smartphone	0.059	0.729	11.8
Pasta 500g →Instant Noodles, Pasta Sauce	0.055	0.693	11.8
Instant Noodles, Pasta Sauce →Pasta 500g	0.055	0.929	11.8
Phone Case, Screen Guard →Smartphone	0.059	0.977	11.8
Smartphone →Phone Case, Screen Guard	0.059	0.717	11.8

Minimum confidence	Rules	Rules with lift >2
30%	347	347
50%	226	226
70%	190	190
90%	107	107

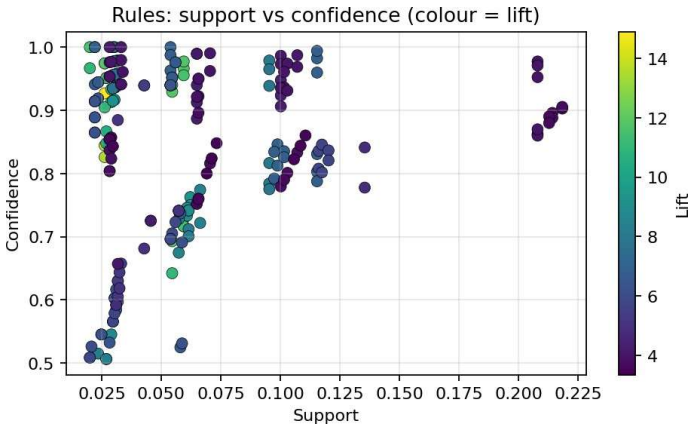


Figure 3: Rules plotted by support and confidence; colour shows lift.

Verification of a rule by direct counting: the top rule was re-computed by counting baskets directly in the database; support, confidence and lift matched the table exactly (see notebook, Section 11).

5. Observations

- **Correct and consistent:** Apriori and FP-Growth find exactly the same frequent itemsets; direct counting confirms the supports.
- **Strong associations:** the strongest rules (lift 11–15) link accessory bundles for smartphones (phone case, screen guard, charger) and pasta-night items (pasta, noodles, sauce). Breakfast items (bread, butter, jam) are the most frequent pair/triple. These agree with how the simulated baskets were generated, showing that the algorithms recover real structure.
- **Lift versus confidence:** a rule can have high confidence only because its consequent is popular; lift corrects for this, so rules should be ranked by lift together with support.
- **Thresholdsmatter:**loweringsupportorconfidencegivesmanymoreandmoretrivialrules;thresholds should be set with business knowledge.
- **Business use:** place associated items close together, create bundle offers (for example phone case + screen guard with a smartphone) and plan promotions. Closed and maximal itemsets give a compact representation of the same knowledge.
- **Limitations:**simulateddata;onlyitempresence(notquantityorprice)isused;support/confidencecan miss rare but valuable associations; no time or customer information is used.

6. Conclusion

Association rule mining was implemented end to end: data extraction from the warehouse, preprocessing, two algorithms written from scratch, rule generation and evaluation with five metrics. FP-Growth gave the same results as Apriori with less work, and lift identified the most meaningful product associations among 226 candidate rules.

References

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